School of Engineering Mathematics and Technology

SEMT20001: Principles of Computational Modelling

COURSEWORK ASSESSMENT: QUESTION 2

The question below forms one part of the assessment for the unit.
This is question 2 of 4.
Each question carries 25 marks. Answer all questions.
The maximum mark for this assessment is 100 marks.
The deadline for submission for all questions is **1pm on 19 March 2026**.

You must submit **two** items (on Blackboard) for **each** question:

(1) a single self-contained Jupyter Notebook, that reports your answers to the question, and explains how you found them, including all relevant Python commands and functions, and explanations of your code design, **and**

(2) a single .zip file containing all Python .py files.

Your Jupyter Notebook (1) for this question must contain everything you wish to be marked for this question (including, but not limited to, numerical and textual answers, figures, and all Python code used to generate them). Material not contained in your document (1) will receive no credit. Any Python .py files that are not contained in your .zip file (2) will receive no credit.

Your submission must run from the `pocm2526` environment, as defined on Blackboard (see *Learning and Teaching Area : Read me first! Python starter guide* for details). Any code that does not will receive no credit.

Your submission must be your own individual effort; submissions will be checked for possible plagiarism. You may only use tools such as spelling and grammar checkers in this assignment, and their use should be limited to corrections of your own work rather than substantial re-writes or extended contributions. Use of AI applications (even if they are free) beyond this is contract cheating; see `https://www.bristol.ac.uk/students/support/academic-advice/academic-integrity/contract-cheating/`.

Question 2: Discrete vs continuous

(a) Solve the boundary value problem

$$t^2\ddot{x}(t) + t\dot{x}(t) + (t^2 - 1)x(t) = 0,$$
$$x(-8) = -1$$
$$x(8) = 1$$

for $x : [-8, 8] \to \mathbb{R}$ using Chebyshev collocation on the interval $t \in [-8, 8]$. The exact solution can be written using the Bessel function of the first kind J_α, with $\alpha = 1$, that is

$$x(t) = \frac{J_1(t)}{J_1(8)}.$$

Find the number of collocation points after which the accuracy of the solution does not improve. Illustrate the convergence in a diagram and plot the best possible solution. The required Bessel function can be found in the SciPy library `https://docs.scipy.org/doc/scipy-1.17.0/reference/generated/scipy.special.jv.html`.

(5 marks)

(b) Find all roots of the solution of Q2(a) using validated interval bisection. Create a function that interpolates the solution using Chebyshev polynomials on the interval $[-8, 8]$ and can be used with the `mpmath` library. Join connecting intervals and calculate the width of each interval containing a root. Estimate how many binary digits of accuracy is lost (in double precision floating point representation) when using your Chebyshev interpolator.

(5 marks)

(c) This question is a slighly modified version of Q2(a): one of the boundary conditions has changed.

Solve the boundary value problem

$$\begin{aligned} t^2\ddot{x}(t) + t\dot{x}(t) + (t^2 - 1)x(t) &= 0, \\ x(-8) &= 1 \\ x(8) &= 1 \end{aligned}$$

for $x : [-8, 8] \to \mathbb{R}$ using Chebyshev collocation on the interval $t \in [-8, 8]$. The exact solution can be written using Bessel functions of the first J_α and second kind Y_α, with $\alpha = 1$, that is

$$x(t) = \operatorname{Re} \frac{(Y_1(-8) - Y_1(8))J_1(t) + 2J_1(8)Y_1(t)}{J_1(8)(Y_1(-8) + Y_1(8))}.$$

Find the number of collocation points after which the accuracy of the solution does not improve, but do not use more than 1024 collocation points. Illustrate the convergence in a diagram and plot the best possible solution. Summarise your observations and insight about the differential equation. The required Bessel function of second kind can be found in the SciPy library `https://docs.scipy.org/doc/scipy-1.17.0/reference/generated/scipy.special.yv.html`.

(5 marks)

(d) This question is another small modification of Q2(a): it makes the boundary condition nonlinear.

Solve the boundary value problem

$$\begin{aligned} t^2\ddot{x}(t) + t\dot{x}(t) + (t^2 - 1)x(t) &= 0, \\ x(-8) + 0.1x^3(-8) &= -1 \\ x(8) &= 1 \end{aligned}$$

for $x : [-8, 8] \to \mathbb{R}$ using Chebyshev collocation on the interval $t \in [-8, 8]$. Use Newton's method to find the solution to the discretised nonlinear equations. This time there is no exact solution.

Choose the solution with $n = 512$ gridpoints as your reference and compare other solutions to it by interpolating the reference solution.

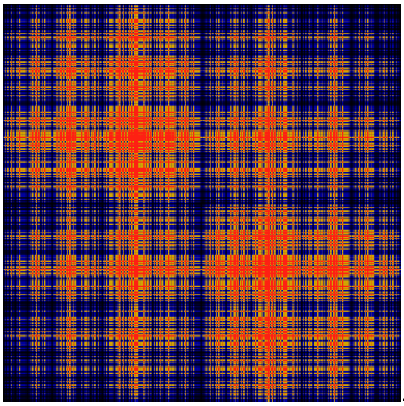
Describe how you can incorporate the nonlinear boundary condition into the discretised equations so that you are able to solve then using Newton's method.

(5 marks)

(e) Consider the Baker's map

$$f(x, y) = \begin{cases} (2x, \frac{y}{2}) & \text{for } 0 \leq x < \frac{1}{2} \\ (2 - 2x, 1 - \frac{y}{2}) & \text{for } 0 \leq x < \frac{1}{2} \end{cases}.$$

It is called the Baker's map because it models how dough is folded while kneading. Using indicator functions on the square $S = [0, 1] \times [0, 1]$ create a transfer matrix and calculate the invariant measure of map f. Note that Wikipedia displays a wrong invariant measure:



You will earn bonus marks if you can tell how to modify f to produce the invariant measure displayed on Wikipedia.

(5 marks)